



Technical Service Bulletin

Technical Service Bulletin: TSB110039

Released Date: 22-Feb-2011

Turbocharger Plumbing Routing Issue

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Warranty Statement

The information in this document has no effect on present warranty coverage or repair practices, nor does it authorize TRP or Campaign actions.

Contents

It has come to the attention of Cummins Inc. that the turbocharger oil supply and coolant vent lines on QSK19 and K19 engines built with water-cooled turbochargers have been found in contact on engines in the field. This contact can result in oil or coolant leaks around the turbocharger. It is recommended that the routing of all the turbocharger plumbing lines be checked at the next service event or regularly scheduled 250 hour maintenance interval. If any contact is found, the lines **must** be adjusted so that they are **not** touching. If the lines are damaged, they **must** be replaced.

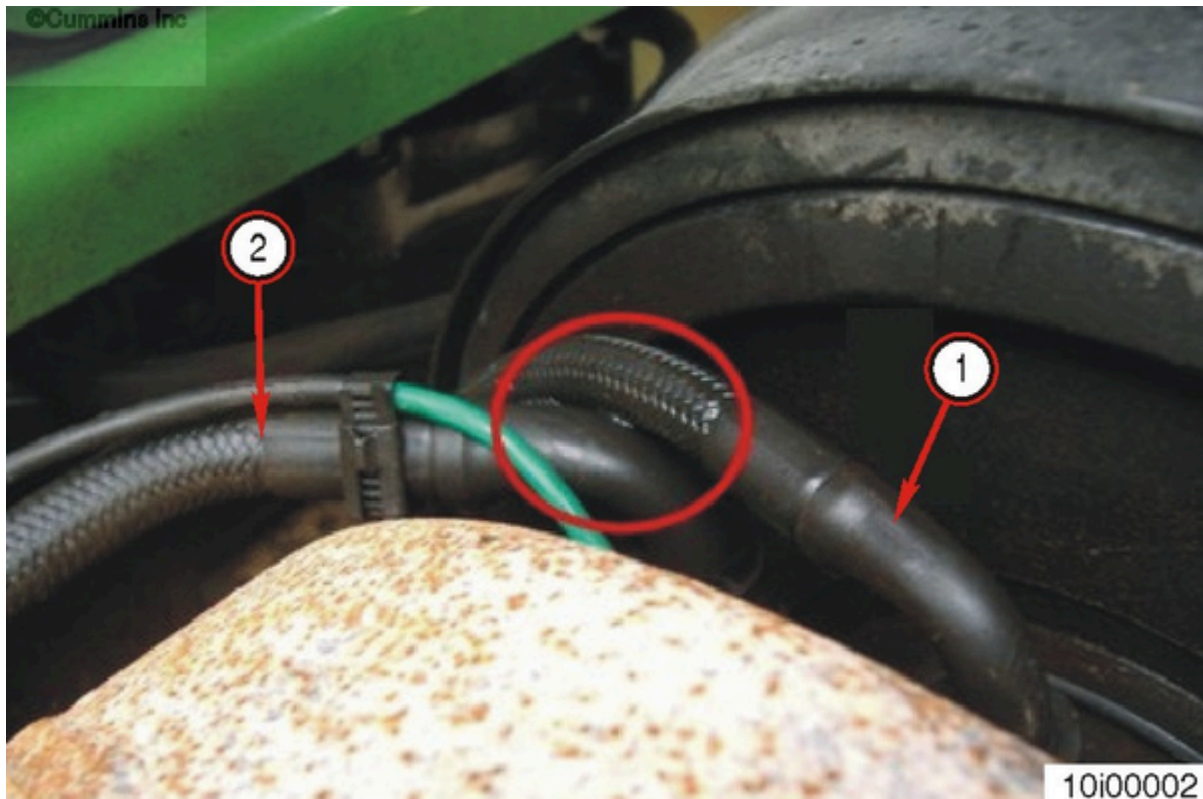


Figure 1 - Oil Supply and Coolant Vent Line Contact

This issue affects all QSK19 and K19 engines built with a water-cooled turbocharger. The most commonly affected area is on top of the turbocharger bearing housing, where the oil supply (1) and coolant vent (2) lines are routed (Figure 1).

This issue is often noticed as coolant or oil leaks around the turbocharger. A leak from the contacted (worn) area on the lines is a common leak path. However, contact also puts stress on the crimped fitting on the end of the lines. This allows for a second leak path at the connection of the braided hose to the crimped fitting.

To verify the issue, check all the turbocharger plumbing lines for contact. Check the lines for evidence of rubbing and replace the lines if any damage is found.

The cause of the contact between the lines is suspected to be improper installation at the plant due to tight clearances around the turbocharger.



Figure 2 - Ideal Routing



Figure 3 - Alternate Routing

The routing of all the turbocharger plumbing lines **must** be checked at the next service event or regularly scheduled 250 hour maintenance interval. If any contact is found, the lines **must** be adjusted so that they are **not** touching. If the lines are damaged, they **must** be replaced. Proper routing is shown in Figures 2 and 3. Ideal routing would be running the lines straight down with no bends to make sure of proper clearances, as shown in Figure 2. However, in some turbocharger configurations, this is **not** possible. If the lines **must** be bent, make sure there is as much clearance as possible between the turbocharger oil line and its surroundings as shown in Figure 3. Make sure the lines are **not** touching the turbocharger housing.

Document History

Date	Details
2011-2-9	Module Created

Last Modified: 22-Feb-2011
